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RAP Meets VS Code

Building Fiori apps with ABAP in a modern IDE

**This demo today runs on an
on-premise ABAP system.**

Training sandbox. Not BTP.

ADT Visual Studio Code Extension.

destination, Claude code Agent.

PART ONE .

Set it up

VS Code

THE ONE EXTENSION

sapse.adt-vscode

ABAP Development Tools for vs code: search the extension marketplace

PART TWO . THE FIRST RELEASE

ADT in VS Code

21 object types, creatable from the editor.

Tables, CDS, behaviour definitions,
classes, service definitions and bindings

CREATE A DATA ELEMENT

Eclipse

DTEL and DOMA are not in the creatable list

There is no embedded SAP GUI.

And none is planned.

Eclipse keeps that job. SE38, SM30, SE16, not here.

And there is no publish command.

For Fiori Apps.

PART THREE .

How the agent knows where to look

**The extension comes
with an MCP server.**

Already running. Nothing to install.

Local HTTP and a bearer token : any MCP client can use it

DEMO

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Install, activate, connect, log on

1. Install sapse.adt-vscode from the marketplace
2. Activate ABAP ADT MCP
3. New ABAP destination → RFC → fill in the connection
4. Log on.
5. Add the destination as a workspace folder
6. Open \$TMP and show a real object opening read-only

On-premise connects over RFC.

HTTPS is for BTP and Public Cloud.

—

A virtual filesystem, not a checkout.

Nothing lands on your disk.

19 tools, in the box

1

Destinations

`abap_list_destinations`

4

Creation

`abap_creation-create_object`

3

Generators

`abap_generators-generate_objects`

2

Business services

`abap_business_services-fetch_services`

5

ATC

`abap_atc_run`

3

Transport

`abap_transport-create`

1

Unit tests

`abap_run_unit_tests`

Additional MCP Server.

Two tools.

THE GAP-FILLING HALF

read_object_source · write_object_source

read and write. No activation tool — deliberately, and never

You can write one in an afternoon.

ADT is a REST API. The agent will write the client for you.

DEMO

~prompts/02-agent.md

Confirm agent can see the system

1. Ask what destinations are connected
2. Confirm the answer is not an empty list
3. Ask it to read one object's source

PART FOUR

Generate the RAP stack and behaviour definitions

prompts/03 and prompts/04

Table yours — the generator reads it

Root view entity the data model, with semantics

Behaviour definition what may happen

Behaviour pool what actually happens — your code

Projection + BDEF what this service exposes

Service definition the contract

Service binding OData V4 — and you publish it

The generator builds from a table.

It never creates one.

So the data model is still your task to do

BUILT BEFORE THIS TALK

ZTTICKETD2

Six data elements, each with a label.

Claude Code

Generate a RAP UI service from table ZTTICKETD2 on the connected destination, into \$TMP.

Managed, with draft. Name the root view ZR_TICKETD4 and the binding ZUI_TICKETD4_04.

prompts/03-generate.md

DEMO

~3 min

One prompt. The whole stack.

1. Paste the generate prompt.
2. Let it run uncut
3. Read the object list
4. Activate the behaviour definition and the pool together

Seven objects, from one sentence.

None of them active.

ACTIVATE

VS Code

but always you, never the agent

**Working CRUD, with a
draft, in one prompt.**

Enforcing absolutely nothing.

Every field editable

Determine and validate

behaviour definition

```
1 determination setInitialValues
2 on modify { create; }
3
4 validation validateStatusTransition
5 on save { field Status; create; update; }
```

CLAS the global shell. Empty here

CLAS_DEF an empty stub. Not what you want

CLAS_IMPL LCL_HANDLER — declaration AND code

CLAS_TEST ABAP Unit

Claude Code

In ZBP_R_TICKETD4, add a validation
validateStatusTransition: a ticket with status
Closed must not go back to New or In Progress.

Declare it in the behaviour definition too,
including under the draft Prepare action.

prompts/04-behave.md

DEMO

Our AI agent writes a validation

1. Paste the validation prompt
2. which source does it write to?
3. Read the generated code before activating anything
4. Activate both halves together

declare the handler

behaviour pool – implementation source

```
1 CLASS lcl_handler DEFINITION
2   INHERITING FROM cl_abap_behavior_handler.
3   PRIVATE SECTION.
4     METHODS validateStatusTransition
5       FOR VALIDATE ON SAVE
6       IMPORTING keys
7       FOR Ticket~validateStatusTransition.
8   ENDCLASS.
```

—

reads the incoming entity

```
1 METHOD validateStatusTransition.  
2 READ ENTITIES OF ZR_TICKETD3 IN LOCAL MODE  
3   ENTITY Ticket FIELDS ( Status )  
4   WITH CORRESPONDING #( keys )  
5   RESULT DATA(tickets).  
6  
7 LOOP AT tickets INTO DATA(ticket).  
8   CHECK ticket-Status = 'C'.  
9 ENDLLOOP.  
10 ENDMETHOD.
```

—

read the stored value

```
1 LOOP AT tickets INTO DATA(ticket).  
2 SELECT SINGLE status FROM ztticketd2  
3   WHERE ticket_uuid = @ticket-TicketUUID  
4   INTO @DATA(stored).  
5  
6 CHECK sy-subrc = 0.  
7 CHECK stored = 'C' AND ticket-Status <> 'C'.  
8 ENDLLOOP.
```

—

failed, and reported

```
1 APPEND VALUE #( %tky = ticket-%tky )
2 TO failed-ticket.
3
4 APPEND VALUE #(
5 %tky = ticket-%tky
6 %element-Status = if_abap_behv=>mk-on
7 %msg = new_message_with_text(
8   severity = if_abap_behv_message=>severity-error
9   text = 'Cannot reopen a closed ticket' ) )
10 TO reported-ticket.
```

And the determination

defaults, so nobody types them

```
1 METHOD setInitialValues.  
2 READ ENTITIES OF ZR_TICKETD3 IN LOCAL MODE  
3   ENTITY Ticket  
4     FIELDS ( Status Priority ReportedOn )  
5     WITH CORRESPONDING #( keys )  
6   RESULT DATA(tickets).  
7 ENDMETHOD.
```

DEMO

~2 min

Close a ticket. Try to reopen it.

1. Activate the behaviour definition and the pool together
2. Refresh the preview
3. Create a ticket — watch the defaults arrive
4. Set it to Closed, save, then try New again
5. Read the error message out loud

NEXT .

Scarfold into a UI5/fiori application

a service is not an app

PUBLISH THE SERVICE BINDING

The Eclipse logo, consisting of the word "Eclipse" in a bold, yellow, sans-serif font, centered within a dark gray rounded rectangle with a thin yellow border.

Eclipse

the extension has no publish command

What the agent never did.

Activate eclipse or vs code by developer

Publish in Eclipse

Start the app developer

Design the data model developer

Thank you

LinkedIn: Victor Fibeckola Okpanachi | Youtube: SAPVIC

Questions

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**Does this work on
S/4HANA private cloud?**

**Yes — same RFC path as
any on-premise system.**

**Everything was \$TMP.
What about transports?**

**Three transport tools ship in the
box. \$TMP is a demo choice.**

**What stops the agent
writing to production?**

**A destination whitelist, a namespace
guard, and no activation tool.**

Where does ATC fit in?

Five tools — run, read, and apply fixes. It is in the box.

Why is there no SAP GUI in VS Code?

It is not planned. Eclipse keeps that job.

Cloud ABAP syntax on a classic system?

The shipped agent file says cloud-compliant. Your box may not be.

Do I still need Eclipse?

Today, yes — for data
elements and for publishing.

What does this cost?

The extension is free. The agent is not.